

Ayush JAIN

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EDUCATION

Carnegie Mellon University

PhD in Robotics

Thesis Advisor : Prof. Katerina Fragkiadaki

Thesis Committee : Prof. Deva Ramanan, Prof. Shubham Tulsiani & Prof. Leonidas Guibas

Partially Supported by :

CMU Robotics Vision Fellowship (24-25 AY)

Meta AI Mentorship Fellowship (25-26 AY)

Pittsburgh, PA

Aug. 2023 – Present

Carnegie Mellon University

Masters in Robotics (4.09 / 4.0)

Thesis Advisor : Prof. Katerina Fragkiadaki

Thesis Committee : Prof. Tom Mitchell, Prof. Shubham Tulsiani & Nikolaos Gkanatsios

Pittsburgh, PA

Aug. 2021 – Aug. 2023

Birla Institute of Technology & Sciences

Bachelor in Computer Science (9.33 / 10.0)

Thesis : Active Embodied Vision - Towards Self-Supervised Never Ending Learners

Thesis Advisor : Prof. Katerina Fragkiadaki & Prof. Pratik Narang

Rajasthan, India

Aug. 2017 – May 2021

POSITIONS

May 2026 August 2026	Google Research Intern, SUNNYVALE, USA I'll be working with Dr. Lan Wang, Prof. Ming-Hsuan Chu and Prof. Leonidas Guibas on 3D Scene Representations and spatial understanding.
May 2025 Nov 2025	Meta Reality Labs Research Scientist Intern, REDMOND, USA Worked with Dr. Adam Harley and Fan Zhang on 4D scene representations and point tracking. It resulted in a first-authored paper currently under submission.
May 2024 Dec. 2024	Meta Facebook AI Research Research Scientist Intern, PITTSBURGH, USA With Dr. Franziska Meier and Dr. Sasha Sax, I worked on building 3D vision-language models. It resulted in 1 first-authored ICML paper and 2 collaborative ICML papers.
August 2021 Present	Carnegie Mellon University Graduate Research Assistant, PITTSBURGH, USA With Prof. Katerina Fragkiadaki, I am working on perception systems static 2D and 3D scenes and robot manipulation.
May 2022 August 2022	Apple Machine Learning Research Research Intern, CUPERTINO, USA With Dr. Navdeep Jaitly and Dr. Miguel Bautista, I worked on few-shot multimodal representation learning.

May 2020 July 2021	Carnegie Mellon University Research Associate, PITTSBURGH, USA Project Page Code Paper With Prof. Katerina Fragkiadaki, I developed a method enabling an embodied agent to learn about objects without ground truth supervision in an unseen 3D environment by allowing the agent to move around.
August 2019 May 2020	MultiCog Research Group Computer Vision Research Assistant, PILANI, India Project Page Code Paper With Prof. Pratik Narang, I built an object detection model for aerial images improving over Retinanet model by 10%
May 2019 August 2019	Indira Gandhi Center for Atomic Research Research Assistant, CHENNAI, India Project Page Code Paper With Dr. N.M. Meenachi, I built the first nuclear language dataset - NText and NQuAD(Nuclear Question Answering Dataset) consisting of 730 question-answer pairs.

PUBLICATIONS

- [14] **Jain, A.**, Paruchuri, S., Gupta, I., Zhang, F., Schmidt, T., Engel, J., Fragkiadaki, K., Harley, A. W., 2026 “TrackEverything : Dense 3D Point Tracking in Long Videos via 3D Scene Representations”. **(In Submission)**
- [13] Lin, A.*, **Jain, A.***, Liu, Y., Fragkiadaki, K., 2026 “Qwen-3D : A Generalist 3D Vision-Language Model for Spatial Understanding”. **(In Submission)**
- [12] Chittupalli, S., **Jain, A.**, Huang, D., 2026 “Pi-HOC : Pairwise 3D Human-Object Contact Estimation”. **(In Submission)**
- [11] **Jain, A.***, Swerdlow, A.*, Wang, Y., Arnaud, S., Martin, A., Sax, A., Meier, F., Fragkiadaki, K., 2025 “Unifying 2D and 3D Vision-Language Understanding”. **(ICML 2025)** [Website](#)
- [10] Sarch, G., Saha, S., Khandelwal, N., **Jain, A.**, Tarr, M. J., Kumar, A., Fragkiadaki, K., 2025 “Grounded Reinforcement Learning for Visual Reasoning.” **(Neurips 2025)** [Website](#)
- [9] McVay, P.*, Arnaud, S.*, Martin, A., Majumdar, A., Jatavallabhula, K. M., Thomas, P., Partsey, R., Dugas, D., Gejji, A., Sax, A., Berges, V.-P., Henaff, M., **Jain, A.**, Cao, A., Prasad, I., Kalakrishnan, M., Rabbat, M., Ballas, N., Assran, M., Maksymets, O., Rajeswaran, A., Meier, F., 2025 “Locate 3D : Real-World Object Localization via Self-Supervised Learning in 3D”.**(ICML 2025)** **(Spotlight)** [Website](#)
- [8] Cao, A., Arnaud, S., Maksymets, O., Yang, J., **Jain, A.**, Yenamandra, S., Martin, A., Berges, V.-P., McVay, P., Partsey, R., Rajeswaran, A., Meier, F., Johnson, J., Park, J. J., Sax, A., 2025 “LIFT-GS : Cross-Scene Render-Supervised Distillation for 3D Language Grounding”. **(ICML 2025)** [Website](#)
- [7] **Jain, A.**, Katara P., Gkanatsios, N., Harley, A., Sarch G., Aggarwal K., Chaudhary V., Fragkiadaki, K., 2024 “ODIN : A Single Model for 2D and 3D Segmentation”. **(CVPR 2024)** **(Highlight)** [Website](#)
- [6] Yang B., Su H., Gkanatsios, N., Ke, T., **Jain A.**, Schneider J., Fragkiadaki, K., 2024 “Diffusion-ES : Gradient-free Planning with Diffusion for Autonomous Driving and Zero-Shot Instruction Following”. **(CVPR 2024)** [Website](#)
- [5] Gkanatsios, N.*, **Jain, A.***, Zhou X., Zhang Y., Atkeson, C., Fragkiadaki, K., 2023 “Energy-based Models are Zero-Shot Planners for Compositional Scene Rearrangement”. **(RSS 2023)** [Website](#)
- [4] **Jain, A.***, Gkanatsios, N.*, Mediratta, I., Fragkiadaki, K., 2022 “Bottom Up Top Down Detection Transformers for Language Grounding in Images and Point Clouds”. **(ECCV 2022)** [Website](#)
- [3] Fang, Z.*, **Jain, A.***, Sarch, G.*, Harley, A., Fragkiadaki, K., 2020 “Move to See Better : Self-Improving Embodied Object Detection”. **(BMVC 2021)** [Paper](#) [Code](#)
- [2] **Jain, A.***, Ramaprasad, R.*, Narang, P., et al., 2020 “AI-Enabled Object Detection in Unmanned Aerial Vehicles for Edge Computing Applications.” **(IEEE Network. 2021)** [Paper](#) [Code](#)
- [1] **Jain, A.**, Meenachi, N.M. and Venkatraman, B., 2020 ”NukeBERT : A Pre-trained language model for Low Resource Nuclear Domain.” **(Arxiv 2020)** [Paper](#) [Code](#)

AWARDS AND SCHOLARSHIPS

2026	Outstanding Reviewer Award, CVPR 2026
2025	Outstanding Reviewer Award, ICCV 2025
2025	Meta-CMU AI Mentorship Fellowship \$150,000 award for 25-26 AY
2024	CMU Robotics Vision Fellowship \$42k award for 24-25 AY
2024	Outstanding Reviewer Award, CVPR 2024
2023	Outstanding Reviewer Award, ICCV 2023
2021-23	Research Scholarship Full tuition funding and stipend for my masters in robotics.
2019	Google AI Summer School 1/50 students across India selected for Google AI summer school
2019	York CVR – VISTA Vision Science Summer School 1/50 students selected worldwide
2018-20	Institute Merit Scholarship Awarded to top 3% students for Exceptional Performance
2017-18	Microsoft Codefundo++ Hackathon Placed 3rd/150+ teams on campus
2016-17	KVPY Scholar A national level scholarship for adjudging high research potential
2016-17	National Science Examination in Physics (NSEP) Fellow

RESEARCH COLLABORATIONS WITH NON-ACADEMIC INSTITUTIONS

Jan 2022	Microsoft Turing Academic Program, REDMOND, USA
July 2023	Website CMU-MSTAP project “Learning instructible visuo-motor agents through multimodal interactive teaching”
Dec 2021	Amazon Alexa Prize SimBot Challenge, SEATTLE, USA
April 2023	Team Page - Developing multimodal instruction following agents as a member of CMU Symbiote Team - Our team got 2nd place in the first phase of the competition.

ACADEMIC AND ADMINISTRATIVE EXPERIENCE

2026	Area Chair for ECCV 2026
2020-Present	Reviewer for Neurips, CVPR, ECCV, ICCV, ICML, ICLR, ICRA, BMVC, AAAI, TPAMI
2024-24	Teaching Assistant, Advanced Computer Vision, CMU
2024-24	Teaching Assistant, Learning for 3D Vision, CMU
2019-20	Teaching Assistant, Artificial Intelligence, BITS PILANI
2019-20	Teaching Assistant, Machine Learning, BITS PILANI
2019-20	Team Leader, Microsoft Student Partner, BITS Pilani
2018-20	Teaching Assistant, Computer Programming, BITS PILANI

TECHNICAL EXPERTISE

Programming	Python , C, C++, Java, HTML, CSS
Frameworks	Pytorch , Tensorflow, Django
Simulators	PyBullet, Ai2Thor, Habitat AI, Open AI Gym

OUTREACH AND VOLUNTEERING

2024-Present	Mentor - ML Collective Office Hours
2024-Present	Mentor - CMU Graduate Application Support Program
2024	Mentor - Robobuddies Mentoring Program
2023-2025	Mentor - CMU Undergrad AI Mentoring Program